Mean / Median /Mode/ Variance /**Standard Deviation** are all very basic but very important concept of statistics used in data science. Almost all the **machine learning algorithm** uses these concepts in data preprocessing steps. These concepts are part of descriptive statistics where we basically used to describe and understand the data for features in Machine learning

Mean :

Mean is also known as average of all the numbers in the data set which is calculated by below equation.

$$\text{Mean} = \frac{\text{Sum of all data values}}{\text{Number of data values}}$$

Symbolically,

$$\bar{x} = \frac{\sum x}{n}$$

where $\bar{x}$ (read as 'x bar') is the mean of the set of x values,
$\sum x$ is the sum of all the x values, and
n is the number of x values.

Lets say we have below heights of persons.

heights=[168,170,150,160,182,140,175,191,152,150]

Mean of dataset

```
In [41]: import numpy as np
         heights=[168,170,150,160,182,140,175,191,152,150]

In [42]: mean=np.mean(heights)
         mean
Out[42]: 163.8
```

Median :

Median is mid value in this ordered data set.

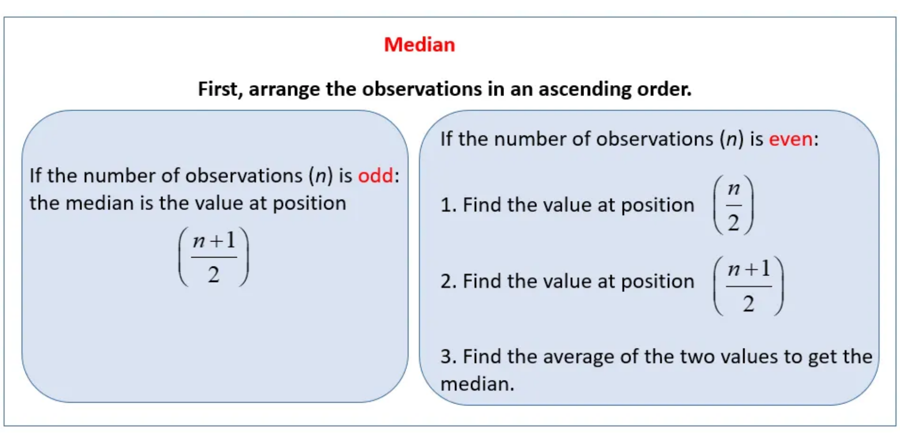

image:source unknown

Arrange the data in the increasing order and then find the mid value.

Sort Data in increasing order

```
In [33]: heights.sort()

In [34]: heights
Out[34]: [140, 150, 150, 152, 160, 168, 170, 175, 182, 191]
```

If we have even number of values in the data set then median is sum of mid two numbers divided by 2

Median of dataset

```
In [37]: median=np.median(heights)

In [38]: median
Out[38]: 164.0
```

In we have odd number in the data set like below we have 9 heights the median will be 5th number value.

```
In [43]: import numpy as np
         heights=[168,170,150,160,182,140,175,191,152]
```

Sort Data in increasing order

```
In [46]: heights.sort()

In [47]: heights
Out[47]: [140, 150, 152, 160, 168, 170, 175, 182, 191]
```

Median of dataset

```
In [44]: median=np.median(heights)

In [45]: median
Out[45]: 168.0
```

Mode :

Mode is the number which occur most often in the data set.Here 150 is occurring twice so this is our mode.

Mode in datset

```
In [49]: import numpy as np
         heights=[168,170,150,160,182,140,175,191,152,150]

In [50]: import statistics as stats
         stats.mode(heights)
Out[50]: 150
```

Variance :

Variance is the numerical values that describe the variability of the observations from its arithmetic mean and denoted by sigma-squared($\sigma 2$)

Variance measure how far individuals in the group are spread out, in the set of data from the mean.

$$\sigma^2 = \frac{\sum_{i=1}^{n} (X_i - \mu)^2}{n}$$

image:Data Analysis in the Geosciences

Where

Xi : Elements in the data set

mu : the population mean

=the population mean

Step 1: This formula says that take each element from dataset(population) and subtract from mean of data set.Later sum all the values.

Step 2: Take the sum in Step 1 and divide by total number of elements.

Square in the above formula will nullify the effect of negative sign(-)

Variance

```
In [51]: import numpy as np
         heights=[168,170,150,160,182,140,175,191,152,150]

In [52]: np.var(heights)
Out[52]: 235.35999999999999
```

Standard Deviation :

It is a measure of dispersion of observation within dataset relative to their mean.It is square root of the variance and denoted by Sigma (σ) .

Standard deviation is expressed in the same unit as the values in the dataset so it measure how much observations of the data set differs from its mean.

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

σ = population standard deviation
N = the size of the population
x_i = each value from the population
μ = the population mean

Standard Deviations

```
In [54]: import numpy as np
         heights=[168,170,150,160,182,140,175,191,152,150]

In [56]: std=np.std(heights)

In [57]: std
Out[57]: 15.341447128612085
```